

**Name : Shan**  
**Sap : 72387**  
**Sem 2**  
**OOP LAB**  
**Tasks**

◆ **LAB Task 8:**

- ✧ **Q1:Write a Java program to demonstrate Compile-time Polymorphism (Method Overloading) using a Calculator class.**

**Input**

**//Calculator Class**

```
package com.mycompany.polymorphism;

public class Calculator {

    // Shan,72387,BSAI
    void sum() {
        int a = 5;
        int b = 6;
        int c = a + b;
        System.out.println("The result is " + c);
    }

    // Shan,72387,BSAI
    void sum(int a, int b) {
        int c = a + b;
        System.out.println("The result is " + c);
    }

    // Shan,72387,BSAI
    void sum(double a, int b) {
        double c = a + b;
        System.out.println("The result is " + c);
    }

    // Shan,72387,BSAI
    void sum(int a, int b, int c) {
        int res = a + b + c;
        System.out.println("The result is " + res);
    }
}
```

## //Polymorphism(Main Class)

```
package com.mycompany.polymorphism;

public class Polymorphism {
    public static void main(String[] args) {

        int a = 10;
        int b = 15;
        int c = 20;

        // Shan,72387,BSAI
        Calculator calc = new Calculator();

        // Shan,72387,BSAI
        calc.sum(a, b, c); // Shan,72387,BSAI
        calc.sum(2.5, 5);
        calc.sum(a, b);
        calc.sum();      // Shan,72387,BSAI
    }
}
```

## Output

```
t - Run (Polymorphism) x
| --- compiler:3.13.0:compile (default-compile) @ Polymorphism ---
Recompiling the module because of changed source code.
Compiling 2 source files with javac [debug release 25] to target\classes

| --- exec:3.5.1:exec (default-cli) @ Polymorphism ---
The result is 45
The result is 7.5
The result is 25
The result is 11

-----
BUILD SUCCESS
-----

Total time: 2.175 s
Finished at: 2026-03-17T11:15:16+05:00
-----
```

- ✧ **Q2:Write a Java program to demonstrate Method Overloading. Create a Registration class with an overloaded method named enroll() that takes different student details as parameters (e.g., only name, name & ID, and name, ID & degree program). Test these methods from a main class.**

## Input

### //Registration Class

```
package com.mycompany.polymorphism;

public class Registration {

    // Shan,72387,BSAI
    void enroll(String name) {
        System.out.println("Student Name: " + name);
    }

    // Shan,72387,BSAI
    void enroll(String name, int id) {
        System.out.println("Student Name: " + name + " | ID: " + id);
    }

    // Shan,72387,BSAI
    void enroll(String name, int id, String program) {
        System.out.println("Student: " + name + " | ID: " + id + " | Program: " +
program);
    }
}
```

### //MainRegistration(Main Class)

```
package com.mycompany.polymorphism;

public class MainRegistration {
    public static void main(String[] args) {

        // Shan,72387,BSAI
        Registration reg = new Registration();

        System.out.println("--- Registration System ---");

        // Shan,72387,BSAI
        reg.enroll("Shan");

        // Shan,72387,BSAI
        reg.enroll("Shan", 72387);

        // Shan,72387,BSAI
        reg.enroll("Shan", 72387, "BSAI");
    }}
}
```

## Output

```

- Run (MainRegistration) x
--- compiler:3.13.0:compile (default-compile) @ Polymorphism ---
Nothing to compile - all classes are up to date.

--- exec:3.5.1:exec (default-cli) @ Polymorphism ---
--- Registration System ---
Student Name: Shan
Student Name: Shan | ID: 72387
Student: Shan | ID: 72387 | Program: BSAI
-----
BUILD SUCCESS
-----
Total time: 1.015 s
Finished at: 2026-03-17T11:36:17+05:00
-----

```

✧ **Q3: Write a Java program to demonstrate Method Overloading. Create a Shape class with overloaded methods named calculateArea(). Implement these methods to calculate the area of a Square (takes 1 integer parameter), a Rectangle (takes 2 integer parameters), and a Circle (takes 1 double parameter). Test all methods from a main class.**

## Input

### //Registration Class

```

package com.mycompany.polymorphism;

public class Shape {

    // Shan,72387,BSAI
    void calculateArea(int side) {
        int area = side * side;
        System.out.println("Area of Square is: " + area);
    }

    // Shan,72387,BSAI
    void calculateArea(int length, int width) {
        int area = length * width;
        System.out.println("Area of Rectangle is: " + area);
    }

    // Shan,72387,BSAI
    void calculateArea(double radius) {
        double area = 3.14 * radius * radius;
        System.out.println("Area of Circle is: " + area);
    }
}

```

## //MainRegistration(Main Class)

```
package com.mycompany.polymorphism;

public class MainShape {
    public static void main(String[] args) {

        // Shan,72387,BSAI
        Shape myShape = new Shape();

        System.out.println("--- Area Calculator ---");

        // Shan,72387,BSAI
        myShape.calculateArea(5);

        // Shan,72387,BSAI
        myShape.calculateArea(4, 6);

        // Shan,72387,BSAI
        myShape.calculateArea(2.5);
    }
}
```

## Output

```
Recompiling the module because of changed source code.
Compiling 6 source files with javac [debug release 25]

--- exec:3.5.1:exec (default-cli) @ Polymorphism ---
--- Area Calculator ---
Area of Square is: 25
Area of Rectangle is: 24
Area of Circle is: 19.625
-----
BUILD SUCCESS
-----
Total time:  2.262 s
Finished at: 2026-03-17T11:48:23+05:00
-----
```

## Github Link:

**<https://github.com/zahidmehar/Shan-72387-OOP-LAB-BSAI.git>**